# Xiaojuan Gou

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## EDUCATION

- **The University of New South Wales**

  Visiting Ph.D. researcher in Biotech & Biomolecular Science — Sydney, Australia

  Supervisor : Associate Professor Emily Wong — *Apr. 2025 – Present*

- **Huazhong Agricultural University**

  Ph.D. Candidate (M.S. – Ph.D. Program) in Neuroscience and Immunology — Wuhan, China

  Supervisor : Prof. Gang Cao & Prof. Xiangru Wang — *Sep. 2020 – Present*

- **Sichuan Agricultural University**

  B.S. in Veterinary Medicine — Sichuan, China

  GPA: 4.06/5 (Rank: 1/220) — *Sep. 2016 – Jun. 2020*

## RESEARCH EXPERIENCE

- **Evolutionary convergence and divergence of hippocampal cytoarchitecture between rodents and primates revealed by single-cell spatial transcriptomics (Primary Collaborative Project)**

  Integrated analysis of single-cell transcriptome and spatial transcriptome data to draw a spatial transcriptome map of the hippocampus of three species.

  *(Jun. 2022 – Jan. 2026)*

- **Transport stress induces neuro-immune dysregulation and exacerbates Mycobacterium tuberculosis infection in mice (PhD Thesis Project)**

  Use single cell and bulk transcriptome data to study the regulatory mechanisms between nerves and immunity.

  *(Apr .2023 – Present)*

- **Strategies to package recombinant Adeno-Associated Virus expressing the N-terminal gasdermin domain for tumor treatment**

  Using single-cell transcriptome data analysis to explain rAAV-P2 treatment activates antitumor immunity and inhibits the growth of triple-negative breast cancer. *(Oct .2020 – Dec.2021)*

- **Single-cell transcriptome landscape and antigen receptor dynamic during SARS-CoV-2 vaccination** *(Feb. 2022 – Jul.2023)*

  Mapping single-cell profiles of immune responses following vaccination with an inactivated COVID-19 vaccine.

- **A single-cell transcriptomic atlas of mouse prefrontal cortex maps dynamics of intratelencephalic neurons during postnatal development**

  Assisted in single-cell RNA-seq analysis for neural circuit research, supporting the construction of spatiotemporal atlases of mouse PFC development.

  *(Feb. 2024 – Present)*

- **Identification of Zebrafish Ortholog for Human EC1.45 Enhancer**

  Identified the zebrafish ortholog of the human EC1.45 enhancer and analyzed its conserved regulatory features using sequence alignment, TF motif analysis, machine learning, and genome synteny.

  *(Jun. 2025 – Present)*

## PUBLICATIONS

Lu, Y., He, W., Huang, X., He, Y., **Gou, X.,** Liu, X., Hu, Z., Xu, W., Rahman, K., Li, S., et al. (2021). Strategies to package recombinant Adeno-Associated Virus expressing the N-terminal gasdermin domain for tumor treatment. ***Nature communications*** 12, 7155. https://doi.org/10.1038/s41467-021-27407-0.

Cao, X., Chen, X., Zhu, Y., **Gou, X.,** Yan, K., Yang, B., Men, D., Liu, L., Zhang, Y.A., and Cao, G. (2023). Single-cell transcriptome landscape and antigen receptor dynamic during SARS-CoV-2 vaccination. ***Genes & diseases*** 10, 1675-1686. https://doi.org/10.1016/j.gendis.2022.08.020.

Fei, T., Dong, Y., **Gou, X.,** Zhang, J., Wang, G., Xu, C., et al. (2026). Evolutionary convergence and divergence of hippocampal cytoarchitecture between rodents and primates revealed by single-cell spatial transcriptomics. ***National Science Review***, https://doi.org/10.1093/nsr/nwaf595.

Sun, L., Zheng, H., Huang, Y., Huang, X., Yan, K., Wang, Z., Yang, L., Yue, Y., **Gou, X.,** Du, G., Wang, Y., Wu, X., Liu, H., Chen, H., Ma, D., Han, Y., Dai, J., Cao, G. (2026). Organization of mouse prefrontal cortex subnetwork revealed by spatial single-cell multiomic analysis of SPIDER-Seq. ***National Science Review***. Accept in principle.

Zheng H., Yan, K., **Gou, X.,** Wang, Z., Yang, L., Huang, Y., Liu, H., Dai, J., Sun, L., Cao, G. (2026). A single-cell transcriptomic atlas of mouse prefrontal cortex maps dynamics of intratelencephalic neurons during postnatal development. ***PLOS Biology***, https://doi.org/10.1371/journal.pbio.3003594.

## RESEARCH INTERESTS

- Data analysis related software development.
- Application of single cell and spatial omics technology in "neuro-immune-infection" systems biology.

## SKILLS

- **Languages:** English (IELTS：6.5).
- **Techniques:** Analysis of a series of omics data such as RNA-sequencing, Single-cell and spatial transcriptomic, Hi-C, ChIP-seq, ATAC-seq, CUT&RUN, etc., Use of common packages in python and R,Linux,Database use.